

1. ATOMIC AND MASS NUMBERS

Z = number of protons (p)

$A = Z + N$ | $N = A - Z$

Neutral atom: $e^- = Z$

Ion with signed charge q : $e^- = Z - q$

For a cation q is positive; for an anion q is negative.

2. RADIATION: ENERGY, FREQUENCY, WAVELENGTH

$E = h\nu$

$c = \nu\lambda$ | $\nu = c/\lambda$

$E = hc/\lambda$

$\lambda = hc/E$ | $\nu = E/h$

$h = 6.626 \times 10^{-34} \text{ J s}$; $c = 3.00 \times 10^8 \text{ m s}^{-1}$; $1 \text{ nm} = 10^{-9} \text{ m}$.

3. BOHR MODEL AND TRANSITIONS

$\Delta E = E_2 - E_1 = h\nu = hc/\lambda$

$mvr = nh/(2\pi) = n\hbar$

Absorption: $E_2 > E_1$ | Emission: $E_2 < E_1$

Use the magnitude $|\Delta E|$ for the photon energy.

4. MATTER WAVES AND UNCERTAINTY

$p = mv$

$\lambda = h/p = h/(mv)$

$\Delta x \Delta p \geq h/(4\pi) = \hbar/2$

If m is fixed: $\Delta p = m \Delta v$

λ is the de Broglie wavelength; Δx and Δp are uncertainties.

5. PHOTOELECTRIC EFFECT

$\nu \geq \nu_0$ | $\phi = h\nu_0$

$h\nu = h\nu_0 + KE_{\max} = \phi + KE_{\max}$

$KE_{\max} = h(\nu - \nu_0) = \frac{1}{2}mv_{\max}^2$

$eV_0 = KE_{\max}$

ν_0 : threshold frequency; ϕ : work function; V_0 : stopping potential.

6. QUANTUM NUMBERS

$n = 1, 2, 3, \dots$

$l = 0, 1, 2, \dots, (n - 1)$

$m_l = -l, \dots, 0, \dots, +l$

$m_s = +\frac{1}{2} \text{ or } -\frac{1}{2}$

$l = 0, 1, 2, 3$ correspond to s, p, d, f respectively.

7. ORBITALS AND ELECTRON CAPACITIES

Subshells in shell $n = n$

Orbitals in a subshell $= 2l + 1$

Orbitals in shell $n = n^2$

Maximum e^- in shell $= 2n^2$

Maximum e^- in subshell $= 2(2l + 1) = 4l + 2$

Maximum e^- per orbital $= 2$

Capacities: s^2, p^6, d^{10}, f^{14} .

8. HYDROGEN EMISSION SPECTRUM

$1/\lambda = R^H (1/n^2f - 1/n^2i)$

$n_i > n_f$ | $R^H = 1.097 \times 10^7 \text{ m}^{-1}$

$\nu = c/\lambda$ | $E = hc/\lambda$

$n_f = 1$ Lyman; 2 Balmer; 3 Paschen; 4 Brackett; 5 Pfund.

9. ELECTRON-FILLING RULES

Lower $(n + l)$ fills first

If $(n + l)$ is equal, lower n fills first

$1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p$

$< 5s < 4d < 5p < 6s < 4f < 5d < 6p < 7s$

Pauli: maximum two opposite-spin electrons per orbital. Hund: singly occupy degenerate orbitals with parallel spins before pairing.

10. USEFUL CONSTANTS AND CONVERSIONS

$\hbar = h/(2\pi)$

$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

$E(\text{eV}) = 1240/\lambda(\text{nm})$

$\lambda(\text{nm}) = 1240/E(\text{eV})$

The eV-nm relation follows from $E = hc/\lambda$.